Infineon Linux Bluetooth® release notes

About this document

Scope and purpose

This document offers a summary of the Bluetooth® firmware release, highlighting known bug fixes and updated specifications specific to the AIROC™ combo chip (CYW4373, CYW43439, CYW43455, CYW43022, CYW55513, and CYW55573/2/1) in the latest Linux release. It also includes information on the proprietary Bluetooth® Stack (AIROC™ Bluetooth® Stack) and related code examples.

Intended audience

This document is intended for anyone who uses the Infineon AIROC™ Wi-Fi & Bluetooth® Combo chips on a Linux host.

Table of contents

About this document ... 1
Table of contents ... 2
1 Overview ... 3
1.1 High level summary ... 3
1.2 Device firmware revision details ... 3
2 Firmware changes ... 4
2.1 CYW4373 ... 4
2.1.1 Bug fixes ... 4
2.1.2 Known issues ... 4
2.2 CYW43439 ... 4
2.2.1 Bug fixes ... 4
2.2.2 Known issues ... 4
2.3 CYW43022 ... 4
2.3.1 Bug fixes ... 4
2.3.2 Known issues ... 4
2.4 CYW55573/2/1 ... 4
2.4.1 Bug fixes ... 4
2.4.2 Known issues ... 4
2.5 CYW43455 ... 4
2.5.1 Bug fixes ... 4
2.5.2 Known issues ... 5
2.6 CYW55513 ... 5
2.7 BTSTACK updates - v3.9.2 ... 5
3 Bluetooth® Linux code examples and supported chips ... 6
3.1 Bluetooth® libraries ... 7
4 Tools and documentation ... 8
Glossary ... 9
Disclaimer ... 11

1 Overview

1.1 High level summary

This document details the new features and bug fixes for CYW4373, CYW43439, CYW43022, CYW43455, CYW55573/2/1, and CYW55513 on the Bluetooth® Linux software release.

This software release includes the following:

- Compliance with the recent Bluetooth® specifications
- Bug fixes
- Bluetooth® Stack updates
- Code example updates

1.2 Device firmware revision details

Table 1 Bluetooth® firmware revision details

Device	Bluetooth® firmware version	BT Version	QDID
CYW4373	001.001.025.0123.0000	5.4	228194
CYW43022	003.002.024.0038.0000	5.4	207748
CYW55573/2/1	001.002.087.0318.0000	5.3	192646
CYW43439	001.003.016.0070.0000	5.4	222970
CYW43455	003.001.025.0196.0000	5.3	183141
CYW55513	001.002.032.0135.0000	5.4	Default: 233875 TX 20 dBm: 241038

Note: AIROC™ Bluetooth® Stack v3.9.2 is validated on the Linux Kernel version 6.6.20 with Raspberry Pi CM4.

2 Firmware changes

This section describes bug fixes and issues that may have an impact on various devices.

2.1 CYW4373

2.1.1 Bug fixes

- SCO quality improvements.

2.1.2 Known issues

- No issues reported during internal sanity tests.

2.2 CYW43439

2.2.1 Bug fixes

- SCO connection stability bug fix.

2.2.2 Known issues

- No issues reported during internal sanity tests.

2.3 CYW43022

2.3.1 Bug fixes

- Boot up stability bug fix

2.3.2 Known issues

- Filter list not maintained after wake from sleep.

2.4 CYW55573/2/1

2.4.1 Bug fixes

- Bug fixes in Bluetooth® LE Stack specific to Bluetooth® 5.2 features.

2.4.2 Known issues

- Bluetooth® ACL disconnection happens when using EV3 on eSCO and 2/3 MBps packet on Bluetooth® ACL as IUT does not respond to 2/3 MBps data packets on Bluetooth® ACL.
- Auracast synchronizer intermittently fails to establish sync to Auracast Broadcaster.

2.5 CYW43455

2.5.1 Bug fixes

- BT 5.3 updates.

2.5.2 Known issues

- No issues reported during internal sanity tests.

2.6 CYW55513

CYW55513, CYW55512, and CYW55511 are designed to address the needs of the Internet of Things (IoT) devices that require minimal power consumption and compact size.

CYW55513/CYW55512/CYW55511 support Bluetooth®/Bluetooth® Low Energy (LE)

- Bluetooth® 5.4 (BDR + EDR + Bluetooth® Low Energy)
- Bluetooth® 5.0/5.1/5.2/5.3/5.4 features:
  - LE long range (LR)
  - LE 2 Mbps, LE 1 Mbps
  - LE mesh
  - Advertising extensions
- LE audio
  - LE audio with LC3 codec on the device (Offload mode)
  - LE audio with LC3 codec on the host (HCI mode)
  - LE isochronous channels

Supported features	Known issues
Coding Scheme Selection for Advertising (CSSA)	None
Periodic advertising ADI support, LE Channel Classification, and LE Enhanced Connection Update	LE CIS connection establishment failure is observed when the corresponding LE-ACL connection is subrated.
LE Core Isochronous channels, LE Power Control	Disconnection due to MIC failure is observed in certain scenarios with multiple LE-CIS links. Disconnection due to connection timeout is observed in certain scenarios with IUT as peripheral in single LE-CIS. Auracast synchronizer intermittently fails to establish sync to Auracast Broadcaster.
Periodic Adv Sync Transfer	None
Extended Advertising Feature, 2 Mbps LE	None
LE Privacy, DPLE	None

2.7 BTSTACK updates - v3.9.2

- Modified `wiced_bt_dev_read_tx_power()` to send the HCI_Read_Transmit_Power_Level HCI command.
- Added a new wiced API, wiced_bt_set_transmit_power_range() (See API documentation for details)
- Added a fix to disallow signed write command on EATT channel as per BT Core Spec.
- Improved Doxygen API content and formatting.

3 Bluetooth® Linux code examples and supported chips

Table 2 Bluetooth Linux code examples and supported chips

Code example	Feature demonstration	Supported chips
Linux Bluetooth® Find me	Demonstrates the Find Me profile, which defines the behavior when a button is pressed on one device to cause an alerting signal on a peer device.	CYW55573/CYW55572/CYW55571, CYW4373, CYW43022, CYW43439, CYW55513.
Linux Bluetooth® hello sensor	Demonstrates GATT database and device configuration initialization, sending data to the client and processing write requests from the client	CYW55573/CYW55572/CYW55571 CYW43439, CYW4373, CYW43022, CYW55513.
Linux Bluetooth® Wi-Fi onboarding	Demonstrates a feature that enables devices to connect to a Wi-Fi access point without requiring a physical interface. This eliminates the need for cumbersome cables and allows for communication of Wi-Fi information through Bluetooth®, including SSID, password, control command, and device Wi-Fi status, providing a true wireless solution.	CYW55573/CYW55572/CYW55571 CYW43439, CYW4373.
Linux Bluetooth® Headset	Multiple profile CE, which demonstrates the use cases and abilities of audio-related functions like A2DP, AVRCP CT, HFP. It also supports GFPS, making it easier for the user to pair devices with as little user interaction as possible.	CYW55573/CYW55572/CYW55571 CYW4373, CYW43022, CYW43439, CYW55513.
Linux Bluetooth® Wake on LE	Demonstrates use of APCF function to achieve Wake on LE.	CYW55573/CYW55572/CYW55571, CYW55513.
Linux Bluetooth® LE Audio CIS Source	Implement the unicast source application using BTSTACK and LE Audio profile library.	CYW55573/CYW55572/CYW55571, CYW55513.
Linux Bluetooth® LE Audio CIS Sink	Implement the unicast sink application using BTSTACK and LE Audio profile library.	CYW55573/CYW55572/CYW55571, CYW55513.
Linux Bluetooth® LE Audio BIS Source	Demonstrates the ability of LE Audio broadcast.	CYW55573/CYW55572/CYW55571, CYW55513.
Linux Bluetooth® LE Audio BIS Sink	Demonstrates the ability to receive LE Audio broadcast.	CYW55573/CYW55572/CYW55571, CYW55513.

3.1 Bluetooth® libraries

Table 3 Bluetooth® libraries

Libraries and middleware	Library details
bluetooth-linux	The bluetooth-linux is the adaptation layer (porting layer) between the Linux Bluetooth® application code example and Infineon's BTSTACK running on the Linux-based platforms. The porting layer provides Bluetooth® Stack initialization and implements platform interfaces to provide OS and memory services, enabling communication between the BTSTACK and the Bluetooth® controller.
bt-audio-profiles	Contains source and header files for A2DP, AVRCP, HFP, and SPP profiles.
btsdk-gfps	Contains source and header files for Google Fast Pairing service.
Btstack	BTSTACK is Infineon's Bluetooth® Host Protocol Stack implementation. The stack is optimized to work with Infineon Bluetooth® controllers. The BTSTACK supports Bluetooth® BR/EDR and Bluetooth® LE core protocols.
le-audio-profiles-linux	Provides implementation of various LE Audio profiles, GATT interface utility for LE Audio code example, audio module library, and ISOC data handler interface.
Fw	Folder contains various firmware files for the AIROC™ combo chip (CYW4373, CYW43439, CYW43022, CYW43455, CYW55513 and CYW55573/2/1).

4 Tools and documentation

Please refer to the following documents for more details:

Tools	Scope
Manufacturing Bluetooth® Test tool (MBT)	The manufacturing Bluetooth® Test tool (MBT) is used to test and verify the RF performance of the Infineon Bluetooth® Classic and Bluetooth® Low Energy devices on Linux platforms. See MBT on GitHub.
AIROC™ Bluetooth ® Test and Debug tool	AIROC™ Bluetooth® Test and Debug tool is a GUI tool for testing and debugging Infineon Bluetooth® devices. AIROC™ Bluetooth® Test and Debug Tool connects to the Bluetooth® devices at HCI protocol layer and currently supports HCI UART and HCI USB transport interfaces. The tool allows user to send Bluetooth® HCI commands and receive Bluetooth® HCI events from the Bluetooth® controller of the connected devices. Download the tool from the AIROC™ Bluetooth® Test and Debug webpage.

Glossary

A2DP
advanced audio distribution profile

ACL
asynchronous connection-less

ADI
Advertisement data information

AEC-CCM
advanced encryption standard-counter with cipher block chaining message authentication code

API
application programming interface

APCF
advertising packet content filter

AVRCP
audio/video remote control protocol

BIS
broadcast isochronous stream

BR
basic rate

BTSTACK
Bluetooth® Stack

CIS
connected isochronous stream

CSSA
Coding Scheme Selection for Advertising

EDR
enhanced data rate

GATT
generic attribute profile

HCI
host controller interface

HFP
hands-free profile

ICS
implementation conformance statement

ISOC
isochronous channels

LL
link layer

MOS
mean opinion score

OS
operating system

PAwR
periodic advertising with responses

PDU
protocol data unit

Rx
Reception

SCO
synchronous connection oriented

SMP
security manager protocol

SPP
serial port profile

TCRL
test case reference lists

Tx
transmission

WBS
wide band scheme

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